

Muhammad Shahwaiz

AI Systems & Backend Engineering

Final-year Software Engineering student at UMT Sialkot and co-founder of Menzync. I build Python and TypeScript backends, LLM integrations and automated tests, with a focus on software that shows its evidence, states what it couldn't verify, and leaves important decisions to a person.

EXPERIENCE

AI Intern · FlyRank · Remote Aug 2026 – present

- Audited my recurring work into delegate / collaborate / keep-human categories and set up AI tooling to match.

Software Quality Engineering Intern · Big Brains Aug 2026

Change Makers Ambassador Intern · DataCrumbs May 2026 · 6 weeks

- Community outreach, AI literacy and student engagement.

Data Entry Clerk · Silver Star Enterprises, Daska Jan – Mar 2022

PROJECTS

NextAct **LIVE** · evidence-oriented verification tool nextact.tech

Turns a suspicious message, link, document or payment request into claims, evidence, contradictions, unknowns, a risk level and the safest next step.

- Built case-submission and browser-extension APIs with workspace-scoped database access.
- Wrote SSRF-controlled URL, DNS and TLS evidence collection; adversarial API tests and GitHub Actions checks; Docker/Compose setup.
- Designed AI adapter interfaces with redaction and output validation; risk levels come from a deterministic engine, never the model.

Next.js · TypeScript · pnpm monorepo · Zod · Drizzle · Vitest · Manifest V3 extension

Menzync **RESEARCH + PROTOTYPE** · co-founder & FYP developer NIB IV finalist

- Roman Urdu / English text classification, from a TF-IDF + linear SVM baseline to XLM-RoBERTa, with a Gradio demo.
- Local FastAPI + SQLite prototype with consent-gated saved history, export and deletion; pytest and Vitest tests for account isolation and validation.
- Evaluation and data-audit tooling: macro-F1, per-class metrics, confusion matrices, calibration (ECE, Brier), train/test overlap checks.

EmoSense AI **PROTOTYPE** · CREATE X 2026 AI App Challenge GitHub

- Express backend: three Gemini endpoints constrained by JSON response schemas, request validation, logging, and labelled rule-based fallbacks when the model is unavailable.

Other prototypes

- ClientDesk** (approval-first WhatsApp CRM; FastAPI, PostgreSQL RLS, Celery) · **Aidesk** (document-grounded support with citations and human escalation; FAISS, Gemini)
- DocFlow AI** (retryable document pipeline) · **Beacon** (cited knowledge search; Qdrant, Claude) · **Vantra** (multi-tenant business SaaS)

RECOGNITION

3rd Place, All Pakistan Prompt Engineering Competition (APPEC) 2026 · score 184.97

Finalist, National Idea Bank IV Pre-Incubation 2026 · one of 24 startups selected by ASPIRE Pakistan

Top Performer, Generative AI Application Developer, NCEAC-HEC · 2025

TRAINING

Start Writing Prompts like a Pro · Google via Coursera · 2026

Positive Psychology Specialization Project · University of Pennsylvania via Coursera · 2025

EDUCATION

BS Software Engineering 2023 – present

University of Management and Technology, Sialkot · final year
Deep Learning · NLP · Information Security · Software Quality Engineering · Databases

Intermediate, Pre-Medical 2019 – 2021

Superior College, Daska · Grade A+

SKILLS

Core: Python, TypeScript, FastAPI, Express, Next.js, LLM integration (Claude, Gemini), PostgreSQL, SQLite, pytest, Vitest, Docker

Also used: Celery/Redis, SQLAlchemy/Alembic, Prisma/Drizzle, Qdrant, FAISS, React, Tailwind

Languages: Urdu, Punjabi, English